

Lecture 29

Bayesian Methods for Big Data

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Big data

- ▶ We are living in the era of information coming from numerous sources.
- ▶ The older statistical methods are not anymore easily implementable.
- ▶ This motivated a large number of researchers to build new methods.
- ▶ These include a large number of Bayesian methods as well.
- ▶ We will discuss some of those approaches here.

Sources of big data

- ▶ Data generated from social media
- ▶ Data collected in biomedical and healthcare informatics research such as
 - ▶ DNA sequences
 - ▶ Electronic health records
- ▶ Geospatial data generated by
 - ▶ Remote sensing
 - ▶ Laser scanning
 - ▶ Mobile mapping
 - ▶ Geo-located sensors
 - ▶ Geotagged web contents
 - ▶ Volunteered geographic information (VGI)
 - ▶ Global navigation satellite system (GNSS) tracking
- ▶ and so on...

Different directions of big data research

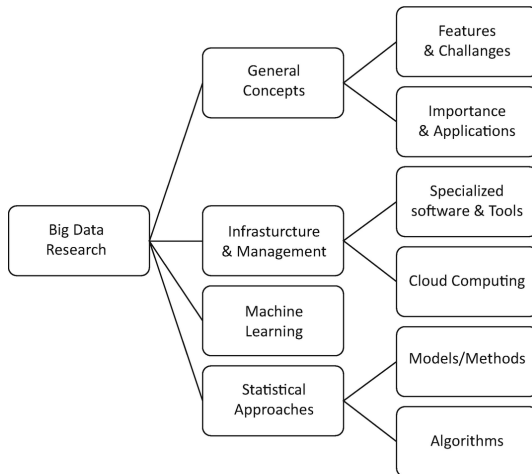


Figure: Source: Jahan et al. (2020)

Statistical Approaches to Big Data

- ▶ The importance of modeling and theoretical results are noted in the literature.
- ▶ Blind trust in algorithms without proper theoretical considerations will not result in valid outputs.
- ▶ Statistical methods play an important role for interpretability, uncertainty quantification, and reducing selection bias.
- ▶ A few important techniques pointed out in the literature are
 - ▶ Confirmatory and exploratory data analysis tools
 - ▶ Data mining methods including supervised and unsupervised learning
 - ▶ Data visualization techniques

Methodological developments

The recent methodological developments for Big Data can be divided into three broad groups

- ▶ *Subsampling*, which calculates a statistic for many subsamples taken from the data and then combine the results
- ▶ *Divide and conquer*, which breaks a dataset into smaller subsets to analyze these in parallel and then combine the results
- ▶ *Online updating of streaming data*, which performs online recursive analytical processing

Four V's of big data

- ▶ Volume
- ▶ Variety
- ▶ Veracity (biases and noise)
- ▶ Velocity

Challenges in Bayesian Approaches for Big Data

Two major challenges are

- ▶ Model construction-related
 - ▶ Models can become complex
 - ▶ A likelihood may not be available
 - ▶ Hierarchical instability
 - ▶ Parameter explosion
 - ▶ Non-identifiability of the model parameters
- ▶ Algorithm development-related
 - ▶ Scalability issues
 - ▶ Computational cost

Bayesian computational methods for Big Data

- ▶ Traditional MCMC do not scale well because they need to iterate through the full data set at each iteration to evaluate the likelihood.
- ▶ A widely used strategy here is the divide-and-conquer sampling by distributing the computation across machines.
- ▶ Here we break a massive dataset into a number of subsets, obtain posterior samples based on each subset in parallel, and finally combine the subset posterior inferences to obtain the full-posterior estimates.
- ▶ The core challenge is the recombination of sub-posterior samples to obtain true posterior samples.

Embarrassingly Parallel MCMC

- ▶ Suppose, we have N IID observations $\mathbf{x}_N = \{x_1, \dots, x_N\}$.
- ▶ For the parameter vector $\theta \in \mathbb{R}^d$, $p(\theta|\mathbf{x}_N) \propto p(\theta)p(\mathbf{x}_N|\theta) = p(\theta) \prod_{i=1}^N p(x_i|\theta)$.
- ▶ We partition $\mathbf{x}_N$ into M subsets $\mathbf{x}^{n_1}, \dots, \mathbf{x}^{n_M}$.
- ▶ For $m = 1, \dots, M$ (in parallel), we sample from the sub-posterior p_m , where

$$p_m(\theta) \propto p(\theta)^{1/M} p(\mathbf{x}^{n_m}|\theta).$$

- ▶ Combine the sub-posterior samples to produce samples from an estimate of the sub-posterior density product $p_1 \times \dots \times p_M$, which is proportional to the full-data posterior, i.e. $p_1 \times \dots \times p_M \propto p(\theta|\mathbf{x}_N)$.

Embarrassingly Parallel MCMC: Example

- ▶ Suppose, we have $X_1, \dots, X_n | \mu \sim \text{Normal}(\mu, \sigma^2)$ IID. Here σ^2 is known.
- ▶ We choose the conjugate prior $\mu \sim \text{Normal}(\mu_0, \sigma_0^2)$.
- ▶ For $m = 1, \dots, M$ (in parallel), we sample from the sub-posterior p_m , where

$$p_m(\mu) \propto p(\mu)^{1/M} p(\mathbf{x}^{n_m} | \mu).$$

- ▶ The posterior distribution for the m -th partition ($n = \sum_{i=1}^M n_m$) is

$$\mu | \mathbf{x}^{n_m} \sim \text{Normal} \left(\mu_P^{(m)} = \frac{\sigma^{-2} \text{sum}(\mathbf{x}^{n_m}) + \mu_0 M^{-1} \sigma_0^{-2}}{n_M \sigma^{-2} + M^{-1} \sigma_0^{-2}}, \sigma_P^{2(m)} = \frac{1}{n_M \sigma^{-2} + M^{-1} \sigma_0^{-2}} \right).$$

- ▶ Thus, $p(\mu | \mathbf{x}_N) \propto p_1 \times \dots \times p_M$.
- ▶ Finally, we have

$$\mu | \mathbf{x}_N \sim \text{Normal} \left(\left[\sum_{m=1}^M \sigma_P^{-2(m)} \right]^{-1} \left[\sum_{m=1}^M \sigma_P^{-2(m)} \mu_P^{(m)} \right], \left[\sum_{m=1}^M \sigma_P^{-2(m)} \right]^{-1} \right).$$

Embarrassingly Parallel MCMC: general case

- ▶ Under suitable regularity conditions, the posterior $p(\theta|\mathbf{x}_N)$ is approximately equal to $\text{Normal}_d(\theta_0, \mathbf{F}_N^{-1})$ (where $\mathbf{F}_N$ is the Fisher information matrix).
- ▶ We thus estimate each sub-posterior with $p_m(\theta) = \text{Normal}_d(\theta|\mu_m, \Sigma_m)$, where μ_m and Σ_m are the sample mean and covariance, respectively, of $\mathbf{x}^{n_m}$.
- ▶ The product of the M sub-posterior densities will be proportional to a Gaussian pdf and our estimate of the density product function $p_1 \times \dots \times p_M$ is

$$\begin{aligned} \hat{p}(\theta|\mathbf{x}_N) &\propto \hat{p}_1 \times \dots \times \hat{p}_M \\ &= \text{Normal}_d \left(\left[\sum_{m=1}^M \Sigma^{(m)-1} \right]^{-1} \left[\sum_{m=1}^M \Sigma^{(m)-1} \mu^{(m)} \right], \left[\sum_{m=1}^M \Sigma^{(m)-1} \right]^{-1} \right). \end{aligned}$$

Robust and scalable Bayes using median posterior

- ▶ Here we apply the divide-and-conquer strategy and then combine the inferences using the median of the subset posteriors.
- ▶ The median posterior (M-posterior) is constructed from the subset posteriors using Weiszfeld's algorithm, which provides a scalable algorithm for robust estimation.
- ▶ Weiszfeld's algorithm is one of the well-known and computationally efficient ways to find the geometric median in Hilbert spaces.
- ▶ For a given set of m points $\mathbf{x}_1, \dots, \mathbf{x}_n$ with each $\mathbf{x}_i \in \mathbb{R}^d$, the geometric median is defined as $\arg \min_{\mathbf{y} \in \mathbb{R}^d} \sum_{i=1}^n \|\mathbf{y} - \mathbf{x}_i\|_2$.
- ▶ For more details, please refer to *Robust and Scalable Bayes via a Median of Subset Posterior Measures* by Minsker et al. (2017), JMLR.

An adaptive subsampling approach

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MH( $p(x|\theta)$ ,  $p(\theta)$ ,  $q(\theta'|\theta)$ ,  $\theta_0$ ,  $N_{\text{iter}}$ ,  $\mathcal{X}$ )  
  1  for  $k \leftarrow 1$  to  $N_{\text{iter}}$   
  2     $\theta \leftarrow \theta_{k-1}$   
  3     $\theta' \sim q(\cdot|\theta)$ ,  $u \sim \mathcal{U}_{(0,1)}$ ,  
  4     $\psi(u, \theta, \theta') \leftarrow \frac{1}{n} \log \left[ u \frac{p(\theta)q(\theta'|\theta)}{p(\theta')q(\theta|\theta')} \right]$   
  5     $\Lambda_n(\theta, \theta') \leftarrow \frac{1}{n} \sum_{i=1}^n \log \left[ \frac{p(x_i|\theta')}{p(x_i|\theta)} \right]$   
  6    if  $\Lambda_n(\theta, \theta') > \psi(u, \theta, \theta')$   
  7       $\theta_k \leftarrow \theta' \triangleright$  Accept  
  8    else  $\theta_k \leftarrow \theta \triangleright$  Reject  
  9  return  $(\theta_k)_{k=1, \dots, N_{\text{iter}}}$ 
```

Figure: Source: Bardenet et al. (2014)

An algorithm for massive online streaming data

- ▶ We consider the case of a simple linear regression model $y = \mathbf{x}'\boldsymbol{\beta} + \varepsilon$, where $\varepsilon \sim \text{Normal}(0, \sigma^2)$.
- ▶ Suppose we choose the standard conjugate priors $\boldsymbol{\beta} \sim \text{Normal}_P(\mathbf{0}, \mathbf{I}_P)$ and $\sigma^2 \sim \text{Inverse-Gamma}(a, b)$.
- ▶ Let $\mathbf{D}^{(t)}$ be denoting the data obtained until time t .
- ▶ The full conditional posterior distributions are

$$\sigma^2 | \boldsymbol{\beta}, \mathbf{D}^{(t)} \sim \text{Inverse-Gamma}(a_t, b_t), \quad \boldsymbol{\beta} | \sigma^2, \mathbf{D}^{(t)} \sim \text{Normal}_P(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t),$$

where a_t , b_t , $\boldsymbol{\mu}_t$, and $\boldsymbol{\Sigma}_t$ follow a recursive pattern (check) in terms of sufficient statistics.

- ▶ Thus, we do not need to calculate the whole data likelihood repeatedly.
- ▶ For more details, refer to *Bayesian Conditional Density Filtering* by Qamar et al. (2015), arXiv.

Different directions of Bayesian modeling research

- ▶ Variable selection in Big Data
- ▶ Bayesian nonparametrics using various stochastic processes like
 - ▶ Gaussian processes
 - ▶ Dirichlet processes
 - ▶ Indian buffet process
 - ▶ Infinite hidden Markov models
- ▶ Bayesian Networks
- ▶ Bayesian Empirical Likelihood

Thank you!